

1. Administrative & Study Identification

Full Study Title: A Phase Ib/II Dose Finding Study Assessing Safety and Efficacy of [177Lu]Lu-DOTA-TATE in Newly Diagnosed Extensive Stage Small Cell Lung Cancer (ES-SCLC) in Combination with Carboplatin, Etoposide and Atezolizumab in Induction and with Atezolizumab in Maintenance Phase **Short Title/Acronym:** A Study of [177Lu]Lu-DOTA-TATE in Newly Diagnosed ES-SCLC Patients in Combination With Carboplatin, Etoposide and Atezolizumab **Protocol Number:** CAAA601A42101 **NCT Number:** NCT05142696 **Posting ID:** 882390602122054608 **Sponsor/Company Name:** Novartis Pharmaceuticals (Novartis) **Trial Status:** RECRUITING **Investigational Product:** [177Lu]Lu-DOTA-TATE (Lutathera) combined with Atezolizumab (Trade Name: Tecentriq, ATC Code: L01FF05, EMA URL: https://www.ema.europa.eu/en/documents/product-information/tecentriq-epar-product-information_en.pdf), Carboplatin, and Etoposide **Phase of Development:** PHASE1, PHASE2 (Phase Ib/II) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: * *Phase Ib:* To establish a safe and well-tolerated dose of [177Lu]Lu-DOTA-TATE in combination with carboplatin, etoposide, and atezolizumab by assessing the frequency of dose-limiting toxicities (DLTs). * *Phase II:* To evaluate the preliminary efficacy of this combination treatment versus the standard combination of carboplatin, etoposide, and atezolizumab alone, measured by Overall Survival (OS). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Objective Response Rate (ORR), Duration of Response (DOR), and overall safety/tolerability (AEs, SAEs).

2.2 Background & Rationale To address the significant medical need for improved therapies in newly diagnosed Extensive Stage Small Cell Lung Cancer (ES-SCLC). ES-SCLC is an aggressive cancer type with limited durable treatment options beyond standard chemoimmunotherapy. This study aims to determine if the addition of targeted radionuclide therapy ([177Lu]Lu-DOTA-TATE), which binds to somatostatin receptors (SSTR) on the tumor, can safely enhance the efficacy and survival outcomes when combined with the standard-of-care atezolizumab, carboplatin, and etoposide. The study will be essential to assess a new potential therapeutic option in participants with this aggressive cancer type.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard-of-care arm: Carboplatin + Etoposide + Atezolizumab alone in Phase II) **Blinding:** Open-label **Randomization:** 1:1 Randomization (Phase II only, following Phase Ib sequential dose escalation with concurrent backfill part)

3.2 Planned Enrollment & Duration Target Enrollment: 200 participants (including dose escalation backfill and approximately 140 Phase II randomized participants) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** Early 2022 **Estimated Primary Completion:** 22-Mar-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant is ≥ 18 years on the day of signing informed consent form. 2. Histologically or cytologically confirmed Extensive Stage Small Cell Lung Cancer (ES-SCLC) with the presence of measurable disease. 3. No prior systemic treatment for ES-SCLC (except the first cycle of chemotherapy with or without atezolizumab of the induction period). 4. ECOG performance status ≤ 1 . 5. Provision of tumor tissue to support exploratory biomarker analysis. 6. Life expectancy of ≥ 6 months.

4.2 Exclusion Criteria 1. Participant has received prior therapy with an antibody or drug against immune checkpoint pathways. 2. Active autoimmune diseases or history of autoimmune diseases that may relapse. 3. Severe chronic or active infections (including active tuberculosis, HBV, or HCV infection) requiring systemic antibacterial, antifungal or antiviral therapy within 2 weeks before Cycle 1 Day 1. 4. Any major surgical procedure requiring general anesthesia ≤ 28 days before Cycle 1 Day 1. 5. History or current diagnosis of electrocardiogram (ECG) abnormalities indicating significant risk of safety for participants participating in the study. 6. Known hypersensitivity to the active substances or any of the excipients of the study drugs. 7. Concurrent participation in another therapeutic clinical study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Induction Period:** [^{177}Lu]Lu-DOTA-TATE (dose declared in Phase Ib) + Atezolizumab (1200 mg Day 1) + Carboplatin (AUC 5 Day 1) + Etoposide (100 mg/m² Days 1-3). Administered every 3 weeks for four cycles. * **Maintenance Period:** [^{177}Lu]Lu-DOTA-TATE + Atezolizumab (1200 mg Day 1) every 3 weeks. * **Note:** The control arm will receive the same induction and maintenance schedule without [^{177}Lu]Lu-DOTA-TATE. **Route of Administration:** Intravenous (IV) Infusion **Duration of Treatment:** 4 cycles of induction followed by continuous

maintenance cycles until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard anti-emetics and supportive care for chemotherapy. **Prohibited:** Concurrent participation in another therapeutic clinical study; systemic antibacterial, antifungal or antiviral treatment within 2 weeks of C1D1.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue Collection, SSTR expression assessment via [68Ga]Ga-DOTA-TATE imaging PET/scan
Baseline	Cycle 1, Day 1	Randomization (Phase II), First Dose of Study Intervention, PK/Blood Radioactivity sampling (Phase Ib)
Interim	Every 3 Weeks	Safety Labs, Efficacy Assessment (CT/MRI), DLT monitoring (Phase Ib), Administration of Treatment Cycles
End of Study	Follow-up	Final Vital Signs, Exit Interview, Follow-up for Disease Progression and Overall Survival

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: * *Phase Ib:* Frequency of dose-limiting toxicities (DLTs), AEs, SAEs, and AEs leading to treatment discontinuation. A dose-limiting toxicity (DLT) is defined as an AE or abnormal laboratory value assessed as unrelated to disease, disease progression, intercurrent illness, or concomitant medications with an onset within the first cycle of initiation of [177Lu]Lu-DOTA-TATE treatment. The National Cancer Institute Common Terminology Criteria for Adverse events (NCI CTCAE) version 5.0 will be used for AE grading. * *Phase II:* Overall Survival (OS), to assess efficacy of [177Lu]Lu-DOTA-TATE in combination with atezolizumab, carboplatin, and etoposide versus standard of care. **Measurement Method:** Clinical safety assessments (Phase Ib); Kaplan-Meier estimation for OS distribution (Phase II).

7.2 Secondary & Exploratory Endpoints * *Phase 1b:* Objective Response Rate (ORR) based on Investigator assessment. * *Phase 1b:* Duration of Response (DOR). * *Phase 1b:* Progression Free Survival (PFS) based on Investigator assessment.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection continuously through the treatment and follow-up phases. Special emphasis on DLT observation during the Phase Ib dose-escalation window.

Serious Adverse Events (SAEs): Reported within 24 hours to the Sponsor/Regulatory agencies as per standard protocol. **Data Monitoring Committee (DMC):** Independent internal/external committee utilized for guiding dose escalation and determining the Recommended Dose (RD) for Phase II.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set (FAS) for efficacy endpoints (OS, PFS, ORR); Safety Set for DLT, AE, and SAE evaluations. **Sample Size Justification:** Target of ~200 total participants. Driven by sequential dose-escalation rules in Phase Ib and adequately powered for 1:1 randomization of ~140 patients in Phase II to detect a clinically meaningful improvement in OS/PFS. **Handling of Missing Data:** Standard survival analysis algorithms utilizing censoring for patients lost to follow-up or alive at data cutoff for time-to-event endpoints (OS, PFS).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of eCRF entries, ensuring protocol adherence, particularly focusing on radiopharmaceutical handling and dosing. **Audit Trail:** Secure EDC database with comprehensive audit trails, eCRF locking, and rigorous data clarification procedures for central and local assessments.

11. Study Identifiers & Governance

Primary ID: CAAA601A42101 **Registry Number:** NCT05142696 **Posting ID:** 882390602122054608 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Study of [177Lu]Lu-DOTA-TATE in Newly Diagnosed ES-SCLC Patients in Combination With Carboplatin, Etoposide and Atezolizumab **Official Regulatory Title:** A Phase Ib/II Dose Finding Study Assessing Safety and Efficacy of [177Lu]Lu-DOTA-TATE in Newly Diagnosed Extensive Stage Small Cell Lung Cancer (ES-SCLC) in Combination With Carboplatin, Etoposide, and Atezolizumab in Induction and With Atezolizumab in Maintenance Phase

12. Clinical Framework (Targeted Radionuclide Specific)

Disease Area / Primary Condition: Extensive Stage Small Cell Lung Cancer (ES-SCLC) **Preferred Name:** Small cell lung cancer extensive stage **Condition Definition:** Small cell lung carcinoma which has spread beyond one hemi-thorax and the regional lymph nodes. **MeSH**

Code: D009369 **SNOMED ID:** 708064003 **SNOMED Hierarchy:** 188554009 | Extensive stage small cell lung carcinoma | Extensive stage small cell lung carcinoma (disorder) **Diagnosis Threshold:** Histologically or cytologically confirmed ES-SCLC with measurable disease **Medical Methodology:** Targeted Radionuclide Therapy combined with Chemoimmunotherapy **Optimization Target:** Identifying the optimal, safe, and tolerable dose of [177Lu]Lu-DOTA-TATE for subsequent Phase II efficacy validation

13. Study Procedures & Methodology

Management Pathway: Standard oncological clinical pathway combined with radioligand administration **Education Protocol (HCP):** Radiopharmaceutical handling, infusion protocols, and radiation safety training per site **Education Protocol (Patient):** Informed consent details regarding radiation exposure, [68Ga]Ga-DOTA-TATE PET/scan preparatory requirements, and therapy expectations **Quality Audit Mechanism:** Independent central radiological review (if applicable) and stringent DLT safety review committee evaluations

14. Observation & Follow-Up Schedule

Total Duration: Estimated through to primary completion in 2029 **Onsite Visit Cadence:** Coinciding with Day 1 of every 3-week treatment cycle **Remote Monitoring Frequency:** As required for survival follow-up post-treatment discontinuation **Checklist Review Interval:** Regular tumor assessments (e.g., every 6-9 weeks depending on cycle progression) per RECIST 1.1

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				